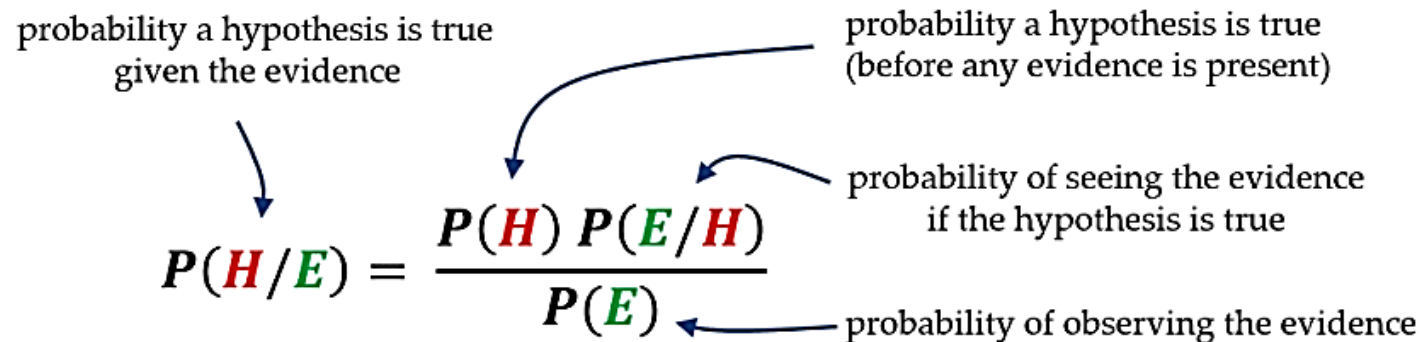


Bayes Rule

In probability theory and statistics, Bayes' theorem (alternatively Bayes' law or Bayes' rule) describes the probability of an event, based on prior knowledge of conditions that might be related to the event.



The diagram shows the Bayes' Rule formula with arrows pointing from descriptive text to parts of the formula:

$$P(\textcolor{red}{H}/\textcolor{green}{E}) = \frac{P(\textcolor{red}{H}) P(\textcolor{green}{E}/\textcolor{red}{H})}{P(\textcolor{green}{E})}$$

- probability a hypothesis is true given the evidence (points to $P(\textcolor{red}{H}/\textcolor{green}{E})$)
- probability a hypothesis is true (before any evidence is present) (points to $P(\textcolor{red}{H})$)
- probability of seeing the evidence if the hypothesis is true (points to $P(\textcolor{green}{E}/\textcolor{red}{H})$)
- probability of observing the evidence (points to $P(\textcolor{green}{E})$)

For example, if cancer is related to age, then, using Bayes' theorem, a person's age can be used to more accurately assess the probability that they have cancer, compared to the assessment of the probability of cancer made without knowledge of the person's age.

$$P(\textit{cause}|\textit{effect}) = \frac{P(\textit{effect}|\textit{cause})P(\textit{cause})}{P(\textit{effect})}$$

Example

Suppose there are three bowls B1, B2, B3.

- Bowl B1 has 2 red and 4 blue coins,
- Bowl B2 has 1 red and 2 blue coins, and
- Bowl B3 contains 5 red and 4 blue coins.

Suppose the probabilities for selecting the bowls is not the same but are:-

- $P(B1) = 1/3$
- $P(B2) = 1/6$
- $P(B3) = 1/2$

Now, let us compute, assuming that a red coin was drawn what will be the probability that it came from bowl B1?

In mathematical terms, we need to find out $P(B1|R)$.

According to Bayes' theorem $P(B1|R) = P(R|B1) * P(B1) / P(R)$

For that first, we need to calculate some probabilities which are:-

Probability to select a red coin i.e $P(R)$

Probability to select the bowl 1 (B1) i.e $P(B1)$ which is already given $1/3$

Probability to select a red coin from B1 i.e $P(R|B1)$

$P(R) = P(B1 \cap R) + P(B2 \cap R) + P(B3 \cap R)$, where,

- $P(B1 \cap R)$ is probability to select bowl 1 and red coin
- $P(B2 \cap R)$ is probability to select bowl 2 and red coin
- $P(B3 \cap R)$ is probability to select bowl 3 and red coin

$P(R) = P(R|B1)P(B1) + P(R|B2)P(B2) + P(R|B3)P(B3)$

$P(R) = P(\text{Number of Red coins} / \text{total number of coins in B1}) * P(\text{selecting B1}) + P(\text{Number of Red coins in B2} / \text{total number of coins in B2}) * P(\text{selecting B2}) + P(\text{Number of Red coins in B3} / \text{total number of coins in B3}) * P(\text{selecting B3})$

$P(R) = 2/6 * 1/3 + 1/3 * 1/6 + 5/9 * 1/2 = 4/9$

$P(R|B1)$ = The probability of selecting a red coin given that it will be drawn from B1 is $2/6$

$P(B1)$ was given i.e. $1/3$.

By putting all the values in formula: $P(B1|R) = (2/6 * 1/3) / 4/9 = 2/8 = 0.25$

So we can say that if a red coin was drawn that it will be 25% chances that it was drawn from bowl 1 i.e. B1.

Naive Bayes Classifier:

- The Naive Bayes classifier calculates the probability of a class C_k given an observation $X = (x_1, x_2, \dots, x_n)$ using Bayes' theorem and the assumption of feature independence. The formula is:

$$P(C_k|X) = \frac{P(X|C_k) \cdot P(C_k)}{P(X)}$$

- Where:
 - $P(C_k|X)$ is the posterior probability of class C_k given observation X .
 - $P(X|C_k)$ is the likelihood of observing X given class C_k .
 - $P(C_k)$ is the prior probability of class C_k .
 - $P(X)$ is the probability of observing X , which acts as a normalizing factor.
- In practice, due to the independence assumption, $P(X|C_k)$ is calculated as the product of the individual conditional probabilities of each feature given the class:

$$P(X|C_k) = P(x_1|C_k) \cdot P(x_2|C_k) \cdot \dots \cdot P(x_n|C_k)$$

- The class with the highest posterior probability is predicted as the class label for the observation X .

The normalizing factor is introduced to scale probabilities so that they sum or integrate to 1, thereby ensuring that they represent valid probability distributions.

Naïve Bayes Classifier

- Naive Bayes is a kind of classifier which uses the Bayes Theorem.
- It predicts membership probabilities for each class such as the probability that given record or data point belongs to a particular class.
- The class with the highest probability is considered as the most likely class. This is also known as **Maximum A Posteriori (MAP)**.

- $\text{MAP}(H) = \max(P(H|E))$
- $\text{MAP}(H) = \max((P(E|H) * P(H)) / P(E))$
- $\text{MAP}(H) = \max(P(E|H) * P(H))$

probability a hypothesis is true given the evidence

probability a hypothesis is true (before any evidence is present)

probability of seeing the evidence if the hypothesis is true

probability of observing the evidence

$$P(H/E) = \frac{P(H) P(E/H)}{P(E)}$$

- $P(E)$ is evidence probability, and it is used to normalize the result. It remains same so, removing it won't affect.
- Naive Bayes classifier assumes that all the features are unrelated to each other. The presence or absence of a feature does not influence the presence or absence of any other feature.
- “A fruit may be considered to be an apple if it is red, round, and about 4” in diameter. Even if these features depend on each other or upon the existence of the other features, a naive Bayes classifier considers all of these properties to independently contribute to the probability that this fruit is an apple.”
- In real datasets, we test a hypothesis given multiple evidence(feature). So, calculations become complicated. To simplify the work, the feature independence approach is used to ‘uncouple’ multiple evidence and treat each as an independent one.
- $P(H | \text{Multiple Evidences}) = P(E1 | H) * P(E2 | H) \dots\dots * P(En | H) * P(H) / P(\text{Multiple Evidences})$

Example

For understanding a theoretical concept, the best procedure is to try it on an example.

Let's consider a training dataset with 1500 records and 3 classes. We presume that there are no missing values in our data. We have 3 classes associated with Animal Types:

- Parrot,
- Dog,
- Fish.

The Predictor features set consists of 4 features as

- Swim
- Wings
- Green Color
- Dangerous Teeth.

Swim, Wings, Green Color, Dangerous Teeth. All the features are categorical variables with either of the 2 values: **T(True)** or **F(False)**.

Example Continued

The below table shows a frequency table of our data. In our training data:

Swim	Wings	Green Color	Dangerous Teeth	Animal Type
50	500/500	400/500	0	Parrot
450/500	0	0	500/500	Dog
500/500	0	100/500	50/500	Fish

- Parrots have 50(10%) value for Swim, i.e., 10% parrot can swim according to our data, 500 out of 500(100%) parrots have wings, 400 out of 500(80%) parrots are Green and 0(0%) parrots have Dangerous Teeth.
- Classes with Animal type Dogs shows that 450 out of 500(90%) can swim, 0(0%) dogs have wings, 0(0%) dogs are of Green color and 500 out of 500(100%) dogs have Dangerous Teeth.
- Classes with Animal type Fishes shows that 500 out of 500(100%) can swim, 0(0%) fishes have wings, 100(20%) fishes are of Green color and 50 out of 500(10%) Fishes have Dangerous Teeth.

Example Continued

Now, it's time to work on predict classes using the Naive Bayes model.

We have taken 2 records that have values in their feature set, but the target variable needs to be predicted.

	Swim	Wings	Green	Teeth
1.	True	False	True	False
2.	True	False	True	True

- We have to predict animal type using the feature values. We have to predict whether the animal is a Dog, a Parrot or a Fish

$$P(H|\text{Multiple Evidences}) = P(E_1|H) * P(E_2|H) * \dots * P(E_n|H) * P(H) / P(\text{Multiple Evidences})$$

Let's consider the first record.

The Evidence here is Swim & Green. The Hypothesis can be an animal type to be Dog, Parrot, Fish.

Example Continued

For Hypothesis testing for the animal to be a Dog:

$$P(\text{Dog} \mid \text{Swim, Green}) = P(\text{Swim} \mid \text{Dog}) * P(\text{Green} \mid \text{Dog}) * P(\text{Dog}) / P(\text{Swim, Green})$$

$$P(\text{Dog} \mid \text{Swim, Green}) = 0.9 * 0 * 0.333 / P(\text{Swim, Green})$$

$$P(\text{Dog} \mid \text{Swim, Green}) = 0$$

For Hypothesis testing for the animal to be a Parrot:

$$P(\text{Parrot} \mid \text{Swim, Green}) = P(\text{Swim} \mid \text{Parrot}) * P(\text{Green} \mid \text{Parrot}) * P(\text{Parrot}) / P(\text{Swim, Green})$$

$$P(\text{Parrot} \mid \text{Swim, Green}) = 0.1 * 0.80 * 0.333 / P(\text{Swim, Green})$$

$$P(\text{Parrot} \mid \text{Swim, Green}) = 0.0264 / P(\text{Swim, Green})$$

For Hypothesis testing for the animal to be a Fish:

$$P(\text{Fish} \mid \text{Swim, Green}) = P(\text{Swim} \mid \text{Fish}) * P(\text{Green} \mid \text{Fish}) * P(\text{Fish}) / P(\text{Swim, Green})$$

$$P(\text{Fish} \mid \text{Swim, Green}) = 1 * 0.2 * 0.333 / P(\text{Swim, Green})$$

$$P(\text{Fish} \mid \text{Swim, Green}) = 0.0666 / P(\text{Swim, Green})$$

The denominator of all the above calculations is same i.e, $P(\text{Swim, Green})$. The value of $P(\text{Fish} \mid \text{Swim, Green})$ is greater than $P(\text{Parrot} \mid \text{Swim, Green})$.

Using Naive Bayes, we can predict that the class of this record is Fish.

Example Continued

Let's consider the second record.

The Evidence here is Swim, Green & Teeth. The Hypothesis can be an animal type to be Dog, Parrot, Fish.

For Hypothesis testing for the animal to be a Dog:

$$P(\text{Dog} \mid \text{Swim, Green, Teeth}) = P(\text{Swim} \mid \text{Dog}) * P(\text{Green} \mid \text{Dog}) * P(\text{Teeth} \mid \text{Dog}) * P(\text{Dog}) / P(\text{Swim, Green, Teeth})$$

$$P(\text{Dog} \mid \text{Swim, Green, Teeth}) = 0.9 * 0 * 1 * 0.333 / P(\text{Swim, Green, Teeth}) = 0$$

For Hypothesis testing for the animal to be a Parrot:

$$P(\text{Parrot} \mid \text{Swim, Green, Teeth}) = P(\text{Swim} \mid \text{Parrot}) * P(\text{Green} \mid \text{Parrot}) * P(\text{Teeth} \mid \text{Parrot}) * P(\text{Parrot}) / P(\text{Swim, Green, Teeth})$$

$$P(\text{Parrot} \mid \text{Swim, Green, Teeth}) = 0.1 * 0.80 * 0 * 0.333 / P(\text{Swim, Green, Teeth})$$

$$P(\text{Parrot} \mid \text{Swim, Green, Teeth}) = 0$$

For Hypothesis testing for the animal to be a Fish:

$$P(\text{Fish} \mid \text{Swim, Green, Teeth}) = P(\text{Swim} \mid \text{Fish}) * P(\text{Green} \mid \text{Fish}) * P(\text{Teeth} \mid \text{Fish}) * P(\text{Fish}) / P(\text{Swim, Green, Teeth})$$

$$P(\text{Fish} \mid \text{Swim, Green, Teeth}) = 1 * 0.2 * 0.1 * 0.333 / P(\text{Swim, Green, Teeth})$$

$$P(\text{Fish} \mid \text{Swim, Green, Teeth}) = 0.00666 / P(\text{Swim, Green, Teeth})$$

The value of $P(\text{Fish} \mid \text{Swim, Green, Teeth})$ is the only positive value greater than 0.

Using Naive Bayes, we can predict that the class of this record is **Fish**.

Advantages/Disadvantages of Naïve Bayes

- Advantages:
 - Fast to train (single scan). Fast to classify
 - Not sensitive to irrelevant features
 - Handles real and discrete data
 - Handles streaming data well
- Disadvantages:
 - Assumes independence of features